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MILITARY SPECIFICATION

BOOSTER, GUIDED MISSILE, MARK 12 MOD 0
(LOADED ASSEMBLY)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of solid propellant booster designated as MK 12 MOD 0.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

	MIL-STD-129	Marking for Shipment and Storage
	MIL-STD-286	Propellants, Solid; Sampling, Examination and Testing
*	MIL-STD-167-1	Mechanical Vibrations of Shipboard Equipment
	MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (501), Indian Head, MD 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1336

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DRAWINGS

Naval Sea Systems Command (Code Ident 10001)

LD 487617	Booster, Guided Missile (Loaded) Assembly, Mark 12 MOD 0 List of Drawings and Specifications (U)
LD 487621	Propellant Grain, Guided Missile, Mark 53 MOD 0 List of Drawings and Specifications (U)
1636730	Booster, Guided Missile, Mark 12 MOD 0 Ballistic Data (U)

PUBLICATIONS

Naval Sea Systems Command (Code Ident 10001)

WR 43	Preparation of Quality Assurance Provisions
WS 12802	Casting Solvent, Propellant
NAVORD Report 6510	Report on the Proper Method of Selecting and Specifying Preservation, Packaging, and Packing
NAVWEPS Form 8821/7	G/M Propulsion Unit Data Sheet
NAVWEPS Form 10342/1	Chemical and Stability Data Sheet

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 100-199	Transportation

(The code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Orders should specify "49 CFR 100-199 (latest revision)").

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Chemical Propulsion Information Agency (CPIA)

CPIA/PP8, Part I — Method for Determining the Tensile Properties of Solid Rocket Propellants

(Application for copies should be addressed to the Chemical Propulsion Information Agency, Applied Physics Laboratory, The Johns Hopkins University, Johns Hopkins Road, Laurel, MD 20810.)

3. REQUIREMENTS

3.1 First article inspection sample.

* 3.1.1 Purpose. The purpose of the first article is to ascertain that the contractor is capable of producing rocket motors which meet this specification, prior to starting production.

* 3.1.2 First article inspection. Unless otherwise specified in the contract (see 6.2), each contractor shall submit a first article of 15 full-scale loaded rocket motors to the procuring activity. These units shall be manufactured using materials, processes, and equipment proposed for production. When the results of the first article (see 4.4) are acceptable, the contractor shall receive written approval of the first article from the procuring activity prior to starting production. Further manufacture of the rocket motor prior to the approval of the first article by the procuring activity shall be at the contractor's risk.

* 3.1.3 Change of material, material vendors, processes, or equipment. Materials, material vendors, processes, and equipment shall not be changed after acceptance of the first article without prior written approval of the procuring activity. A first article may be required by the procuring activity when supplier undertakes a new contract, purchase order, a new material vendor, or requisition to produce similar material under this specification, or if production has been interrupted. The basis for such a requirement is a change in equipment, personnel, procedures, material vendors, or other changes which, in the opinion of the procuring activity, may adversely affect the end product. The contractor shall be responsible for informing the procuring activity of proposed changes. Any production or resumption of the production prior to the written approval of the new first article by the procuring activity shall be at the contractor's risk.

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3.2 Materials, dimensions, and performance. The materials, dimensions, and performance of the Booster, Guided Missile, Mark 12 MOD 0 shall conform to the requirements of this specification and the requirements specified in the drawings, specifications, and other documents listed in LD 487617 and LD 487621. The casting solvent for the propellant shall conform to WS 12802.

3.3 Data. No data are required by this specification or by applicable documents referenced in Section 2, unless specified in the contract (see 6.2).

3.4 Ballistic performance. The rocket motor ballistic performance shall conform to Drawing 1636730.

3.5 2-Nitrodiphenylamine content. The 2-nitrodiphenylamine (2-NDPA) content of each propellant formulation in the grain shall conform to Drawing 1636730.

3.6 Stability and physical properties (propellant). The stability and physical properties of each propellant formulation in the grain shall conform to Drawing 1636730.

* 3.7 Environmental. The rocket motor shall meet the requirements specified herein during and after exposure to the environmental conditions specified in 4.7.1 through 4.7.4.

* 3.8 Serialization. The serial number and identifying contractor letter metal stamped on the inert booster is a pre-assigned (see 6.2) and controlled number and shall be retained on the loaded booster.

3.9 Workmanship. The boosters shall be uniform in quality and shall be free of chips, dirt, corrosion, burrs, sharp edges, or any other defects that would prevent their use for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.2 Classification of inspections. The inspection of the rocket motor shall be classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

c. Preparation for delivery inspection (see 4.9)

4.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C), and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.4 First article inspection. Unless otherwise specified in the contract, the first article (see 3.1) shall be inspected in accordance with the following:

* 4.4.1 Level of inspection. The first article shall be inspected 100 percent for conformance to the requirements of this specification and the applicable drawings. Failure of a unit to meet the drawing or specification requirements may result in rejection of the sample and disqualification of the contractor.

* 4.4.2 Testing. The first article tests shall be conducted at an activity designated by the procuring activity. First article testing shall consist of all tests specified in 4.4.2.1 through 4.4.2.3 and 4.6 through 4.7.4, as indicated in Table I, using the sequence shown in Table I for each motor.

* 4.4.2.1 Chemical and physical properties test. One of the 15 first article motors shall be utilized for chemical and physical testing as specified in 4.6.2 through 4.6.4.

* 4.4.2.2 Performance test. Unless otherwise specified, 14 units of the first article shall be stabilized (see 6.4.1) and test fired as specified in Table I. Acceptance of performance characteristics shall be determined in accordance with MIL-STD-414, variability unknown, standard deviation method, normal inspection, using an acceptable quality level (AQL) of 0.25 percent defective for both upper and lower specification limits at each temperature.

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TABLE I. First article inspection test matrix.

TEST MOTOR NO.	TEMPERATURE CYCLE(4.7.1)		VIBRATION (4.7.2)		DROP TEST(4.7.3)		THERMAL GRADIENT(4.7.4)		VISUAL INSPECTION	X-RAY	STATIC FIRE TEMPERATURE		
	CYCLE A	CYCLE B	-12°C	49°C	-12°C	49°C	49° to -12°C	-12° to 49°C			-12°C	25°C	49°C
1										A*		B	
2										A	B		
3										A			B
4	A								B				C
5		A							B		C		
6			A						B			C	
7					A				B				C
8							A				RESULTANT TEMP		
9	A			C					D	B,E		F	
10	A					C			D	B,E			F
11		A	C						D	B,E	F		
12		A			C				D	B,E			F
13	A			B		C			D	E	F		
14		A	B		C			D			RESULTANT TEMP		
15	CHEMICAL AND PHYSICAL PROPERTIES TEST SAMPLE										-	-	-

* Letters designate sequence of testing for a given motor

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* 4.4.2.3 First article acceptance. When specified in the contract (see 6.2), results of the inspections and tests shall be forwarded to the procuring activity for review and acceptance. Additional production of the rocket motor prior to written notice of acceptance shall be at the manufacturer's risk.

* 4.5 Quality conformance inspection. The quality conformance inspection shall be as specified in 4.5.1 through 4.5.2.2 and shall consist of those examinations and tests specified in 4.6.

4.5.1 Lot.

4.5.1.1 Production lot. A production lot shall consist of the boosters manufactured on a single production line using only one lot of casting powder in each propellant formulation in the grain.

4.5.1.2 Inspection sublot. Inspection sublots shall consist of a maximum of 25 boosters that were sequentially cast in a single lot.

4.5.2 Sampling and acceptance criteria.

* 4.5.2.1 Sampling for ballistic performance. All motors in an inspection sublot or sublots offered for acceptance shall have complied with all other applicable requirements of 3.2. Two motors from each inspection sublot offered for acceptance shall be selected at random for test firing in accordance with 4.6.2. This sampling shall apply only to the first three (3) sublots. After three (3) consecutive sublots have been accepted per 3.4, sampling shall be one (1) motor per sublot selected at random and fired in accordance with 4.6.2. After five (5) additional consecutive sublots have been accepted (see 3.4), one (1) motor from every other sublot shall be selected for testing. This reduced sampling may be maintained for follow-on contracts upon approval from the procuring activity. In the event a propellant failure results in a rejected sublot or causes a sample to fail to meet the requirements, the contractor shall revert to one (1) sample per sublot until five (5) consecutive acceptable sublots have been fired. Failure of a rocket motor to meet ballistic requirements because of propellant failure shall be cause for rejection of all rocket motors in the sublot from which the reject motor was selected. The remaining motors (motors composing the companion sublot in a reduced sampling plan) may be regrouped for acceptance testing. Failure of an acceptance firing for reasons other than propellant shall be cause for rejection of the sublot if cause for failure is typical of all motors in the sublot and cannot be corrected. If rework is approved, another acceptance test shall be conducted for acceptance of the remainder of the sublot.

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4.5.2.2 Sampling for 2-NDPA content, stability, and physical properties (propellant). One propellant grain shall be selected at random from each production lot for 2-NDPA content, stability, and physical properties. Samples shall be prepared for test as follows:

a. Physical property test specimens. Obtain 4 physical property test specimens from the top, middle, and bottom of each propellant formulation in the grain.

b. 2-NDPA content and stability samples. Obtain the 2-NDPA content and stability samples for each propellant formulation in the grain by preparing a composite sample of each formulation from the shavings obtained during machining of the test specimens of 4.5.2.2.a.

4.6 Inspection methods.

4.6.1 Visual and physical examination.

* 4.6.1.1 Inspection of drawing requirements (M101). The configuration, weight, center of gravity, dimensions, and tolerances of the rocket motor shall be verified for compliance with the applicable drawings.

* 4.6.1.2 Materials, components, and processes inspection. It shall be ascertained by the inspection of records that all components (parts, subassemblies, and materials) have been inspected and found to comply with their respective drawings and specifications, and that all specified manufacturing processes have been followed.

* 4.6.1.3 Marking and workmanship inspection. The rocket motor shall be examined for compliance with the marking and workmanship requirements of 3.8 and 3.9, respectively.

4.6.2 Static firing (M102). Ballistic performance for sublot acceptance shall be determined in accordance with the following:

a. Procedure: The ballistic test sample shall be static fired, within 60 days after submission of the sublot, by an organization designated by the procuring activity (see 6.2). All static firing shall be performed with the booster conditioned at 25° ± 2°C using equipment and procedures approved by the procuring activity.

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b. All test boosters shall be conditioned at the firing temperature for a minimum of 96 hours, and shall be fired within 45 minutes of removal from conditioning. Any booster not fired within the required 45 minutes shall be reconditioned until equilibrium temperature is again reached. Two channels of instrumentation shall be used for both pressure and thrust, and ballistic data reported as the average of the two channels unless one channel is obviously in error.

4.6.2 2-Nitrodiphenylamine content. The 2-NDPA content of each propellant formulation in the grain shall be determined in accordance with method 218.4 of MIL-STD-286 on a sample from 4.5.2.2.b.

4.6.3 Stability. Stability of each propellant formulation in the grain shall be determined in accordance with method 406.1 of MIL-STD-286 on a sample from 4.5.2.2.b.

4.6.4 Physical properties. Physical properties of each propellant formulation in the grain shall be determined in accordance with Part 1 of CPIA/PP8 under the following conditions:

a. The test temperature shall be 25° ± 2°C and the strain rate 0.10 inch per minute per inch.

b. Specimens shall be prepared in accordance with 4.5.2.2.a and shall be tested no later than 72 hours after final machining.

c. Elongation shall be measured from the instant of load application to the point of rupture of the specimen. Rupture is established as that point where the tensile stress is 5 percent less than the maximum which has been reached.

* 4.7 Environmental inspection methods. The rocket motor shall be exposed to the environments specified in 4.7.1 through 4.7.4 to verify compliance with the requirements specified herein. The Government reserves the right to test the rocket motor under any combination of environments that the rocket motor may encounter in its intended application (see 6.1).

* 4.7.1 Temperature cycling (M103). The motors shall be temperature cycled as follows. Motors shall be held at the specified temperatures within ± 3°C until the temperature of the propellant has stabilized (see 6.4.1).

Cycle A 49/-12/49/-12/49/-12/49°C

Cycle B -12/49/-12/49/-12/49/-12°C

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4.7.2 Vibration.

a. The round (see 6.4) shall be mounted so as to simulate the launcher environment and undergo the vibrations of Table II distributed about 1 gravity (g) steady state loading at a 0° launcher elevation. Tolerance values for vibration and shock tests shall be: frequency ± 2 percent, acceleration ± 10 percent, and g ± 5 percent.

TABLE II. Vibration testing.

Launcher Condition	Frequency *, hertz (Hz)	Acceleration, g	Minimum Duration, minutes
Synchronizing	7	3.2	10
Tracking	7	3.2	50
Stationary	In accordance with MIL-STD-167-1, type I.		

*Simple harmonic motion

b. A linear vibration with high frequency shock transients superimposed shall be applied in the transverse direction of both control surface planes at the tail of the round. The linear vibration shall be swept from 20 to 600 Hz and shall produce a 300-pound peak force over the entire frequency range. The high frequency shock transients shall have a repetition rate of not less than 100 Hz and shall produce a 30 g peak transients simultaneously along the longitudinal axis of the round as well as in the control surface planes.

c. A torsional vibration shall be applied about the longitudinal axis at the aft end of the round in an exponential frequency sweep from 20 to 300 Hz capable of producing a 3,000 inch-pound torque over the entire frequency range.

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* 4.7.3 Drop test. The drop shock tests shall be performed on motors packed in a MK 200 MOD 0 or MOD 1 metal shipping container. After conditioning to the temperature specified in Table I, the motor shall be subjected to the following two shocks:

a. Aft end of crate lifted and dropped from a height of 3 feet onto the striking plate.

b. Crate dropped horizontally on the side from a height of 1 foot onto the striking plate.

* 4.7.4 Temperature gradient. The motors designated for temperature gradient testing shall be conditioned to the initial temperature, specified in Table I, for a minimum of 96 hours and then conditioned at the secondary temperature, specified in Table I, for 6 hours immediately prior to static firing.

4.8 Data sheets. A G/M propulsion unit data sheet (NAVWEPS Form 8821/7) shall be required for each loaded booster (see 6.2). Chemical and physical properties and stability data shall be reported on NAVWEPS Form 10342/1 (see 6.2).

4.9 Preparation for delivery inspection. Preservation, packaging, and packing shall be examined for conformance to Section 5.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. As defined in NAVORD Report 6510.

5.1.2 Preservation application. The booster shall have preservation applied as specified in the applicable drawings of LD 487617.

5.2 Packing

5.2.1 Level A. (As defined in NAVORD Report 6510) the booster motors shall be packed in shipping containers as specified by the procuring agency (see 6.2).

5.3 Marking. In addition to any special marking required by the contract, the shipping container shall be marked in accordance with the requirements of MIL-STD-129, 49 CFR \171-178, and as specified in the applicable drawings.

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6. NOTES

6.1 Intended use. The Booster, Guided Missile, Mark 12 MOD 0 is intended for use in the Terrier missile, BT-3, BT-3A, HT-3 and in the Standard Missile (ER).

6.2 Ordering data. Procurement documents should specify, but not be limited to, the following:

6.2.1 Procurement requirements

a. Title, number, and revision letter of this specification.

b. Place of delivery.

c. Place of inspection.

d. The organization designated to perform static firing performance tests (see 4.6.2).

e. Type of shipping container (see 5.2.1).

f. Latest revision of LD and all drawings with latest revisions listed thereon.

g. Request for test results (see 3.3 and 4.4.2.3)

h. First article inspection samples required, if applicable (see 3.1)

i. Assignment of unit serial numbers (see 3.8).

6.2.2 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraph:

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Paragraph	Data Requirements	Applicable DID
4.8	Propulsion Unit Data Sheet	---
4.4.2.3	First Article Test Report	DI-T-4902
4.8	Chemical Properties and Stability Data.	---

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.3 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by DAR 1-323.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.4 Definitions. The following definitions apply to this specification when the terms are used herein.

6.4.1 Temperature stabilization. Temperature stabilization consists of exposure of the rocket motor to the specified temperature for at least 96 consecutive hours.

6.4.2 Round. The term "round" means an assembled missile-booster combination. An inert loaded sustainer or simulated weights for missile components other than the rocket motor may be used during testing.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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